

- Research Question:

How does an increase in temperature influence behavioral and physiological stress responses in Hairy Shore Crabs (*Hemigrapsus oregonensis*) during predator encounters with the European Green Crab (*Carcinus maenas*)?

- Hypotheses:

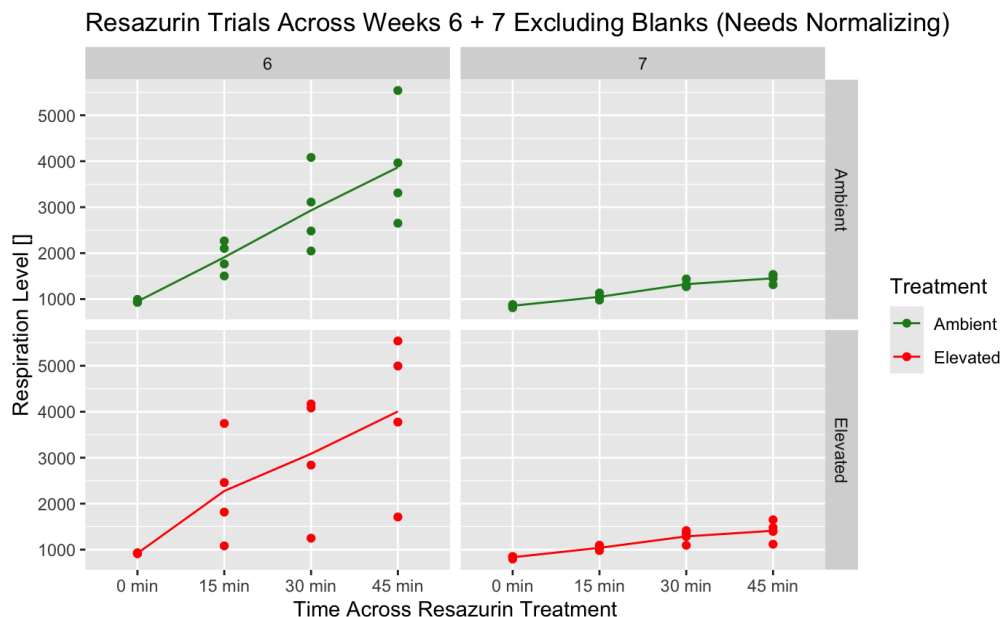
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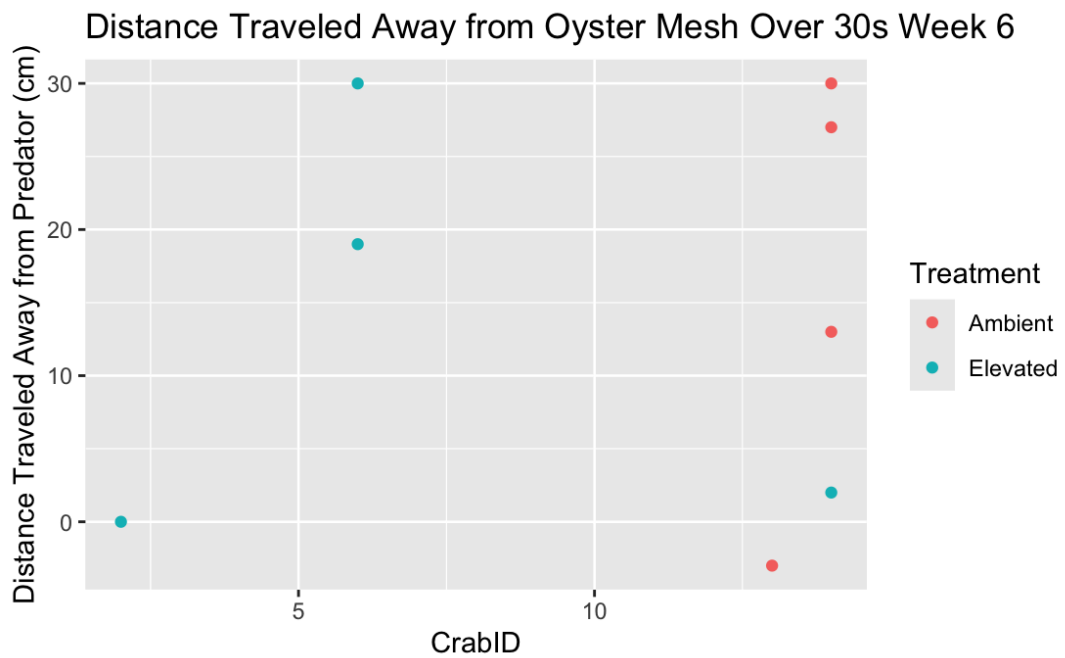
- 1: There will be no significant difference in respiration rate between the two treatments of HSC (*Hemigrapsus oregonensis*)
- 2: There will be no change in righting time and distance travelled away from EGC by the HSC across the two treatments.

Alternative:

- 1: There will be a difference in respiration rates between the two treatments of HSC.
- 2: There will be a change in righting time or distance travelled away from EGC by the HSC across the two treatments.

- Figures:





- Your key takeaway for each figure / result

Crabs in resazurin increased their respiration rate over time, and the rate of increasing was greater in week 6 and less in week 7. There was no significant difference between crabs in ambient temperature water and elevated temperature water.

In the 6th week, ambient crabs had a shorter righting time than elevated crabs. In the 7th and 8th week, elevated crabs had a shorter righting time than the ambient crabs. The difference in values between the ambient and elevated crabs was greater in the 7th week and less in the 8th week.

The third figure is just a preliminary look at one way we will be able to visualize our distance traveled data. I think there's some opportunity to streamline the way the data is entered for easier visualization. The individual dots in a single line show the points where the individual stopped. We can see a clear indication of some crabs preferring to move down the channel and some preferring the mesh.